Predicting 30-Day Hospital Readmissions in Diabetic Patients

Analysis Charts & Visualizations

UCI Diabetes 130-US Hospitals Dataset | April 2026

Final XGBoost Model AUC: 0.690 | 101,766 Encounters | 130 US Hospitals

Figure 1 — Numeric Feature Correlations with 30-Day Readmission

Pearson correlation between each numeric feature and the binary readmission target. number_inpatient is the strongest linear predictor at r = 0.165, followed by number_emergency and number_diagnoses.

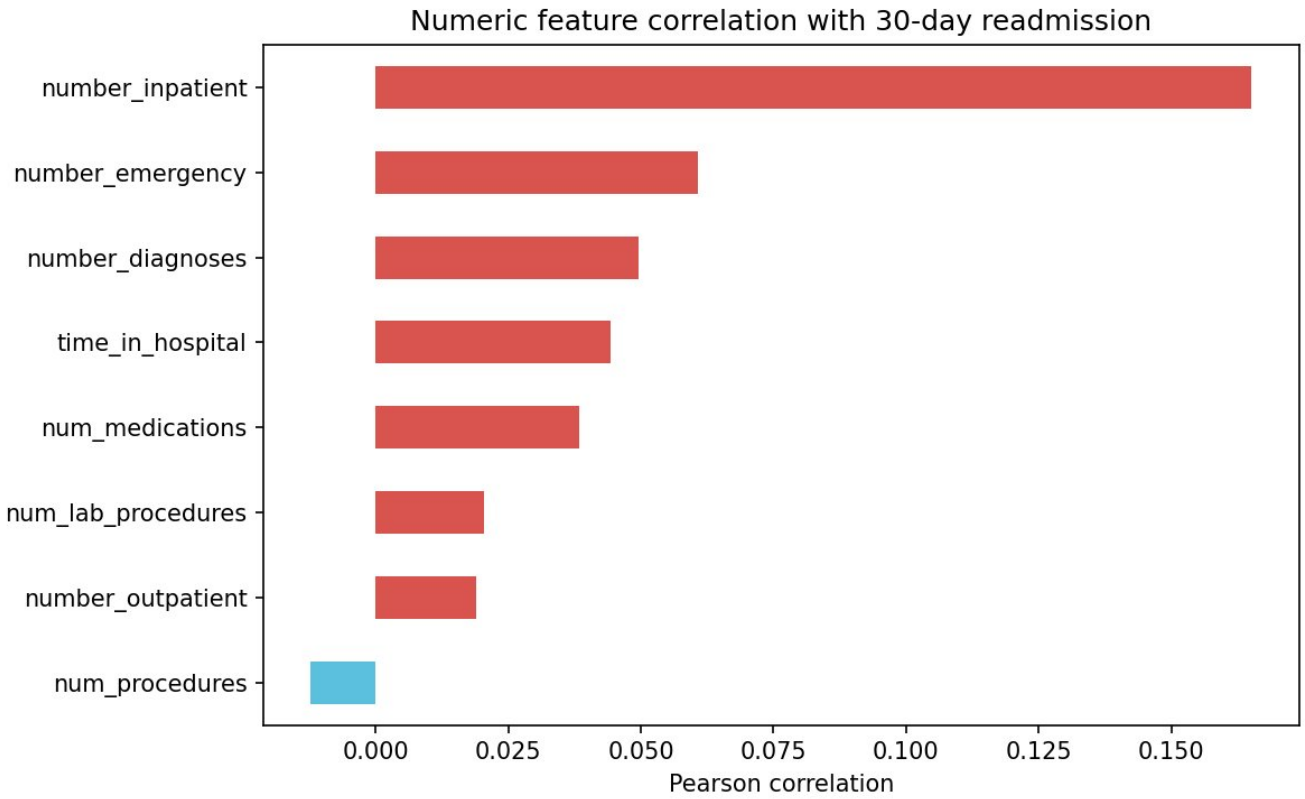


Figure 2 — 30-Day Readmission Rate by Age Group

Readmission rates are relatively flat across age groups (10–12%), with the exception of the small [20-30) bracket. Age was statistically significant (chi-square $p < 0.001$) but contributes modest predictive power.

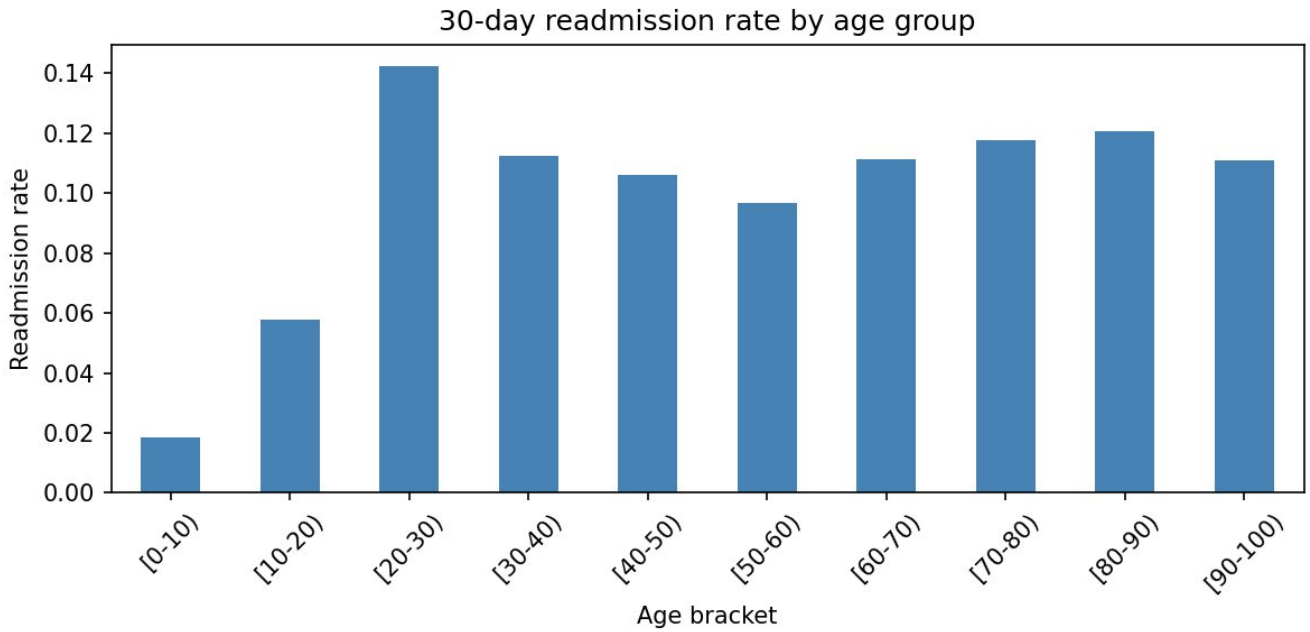


Figure 3 — 30-Day Readmission Rate by Prior Inpatient Visits

The clearest signal in the dataset. Readmission risk rises nearly linearly from 8.5% for patients with no prior inpatient visits to 44% for patients with 8+ prior visits.

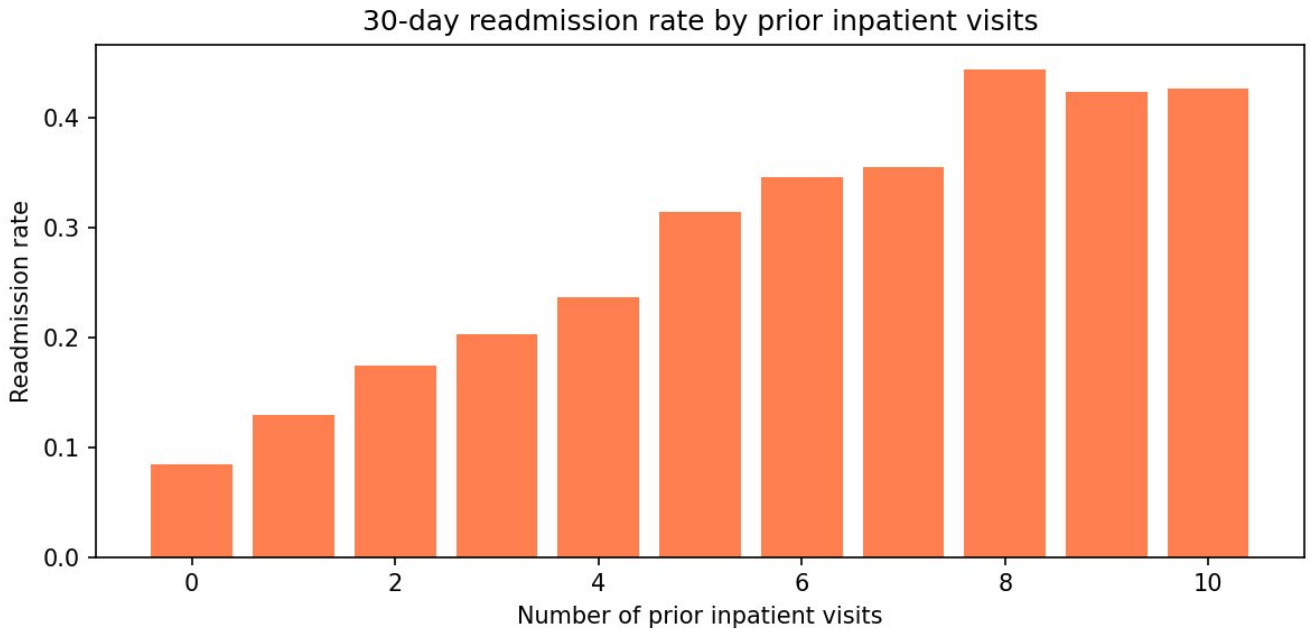


Figure 4 — 30-Day Readmission Rate by Insulin Status

Patients whose insulin was dosed down during the visit had the highest readmission rate (13.9%), followed by those dosed up (13.0%). Patients on steady or no insulin had lower rates, suggesting unresolved glycemic management at discharge.

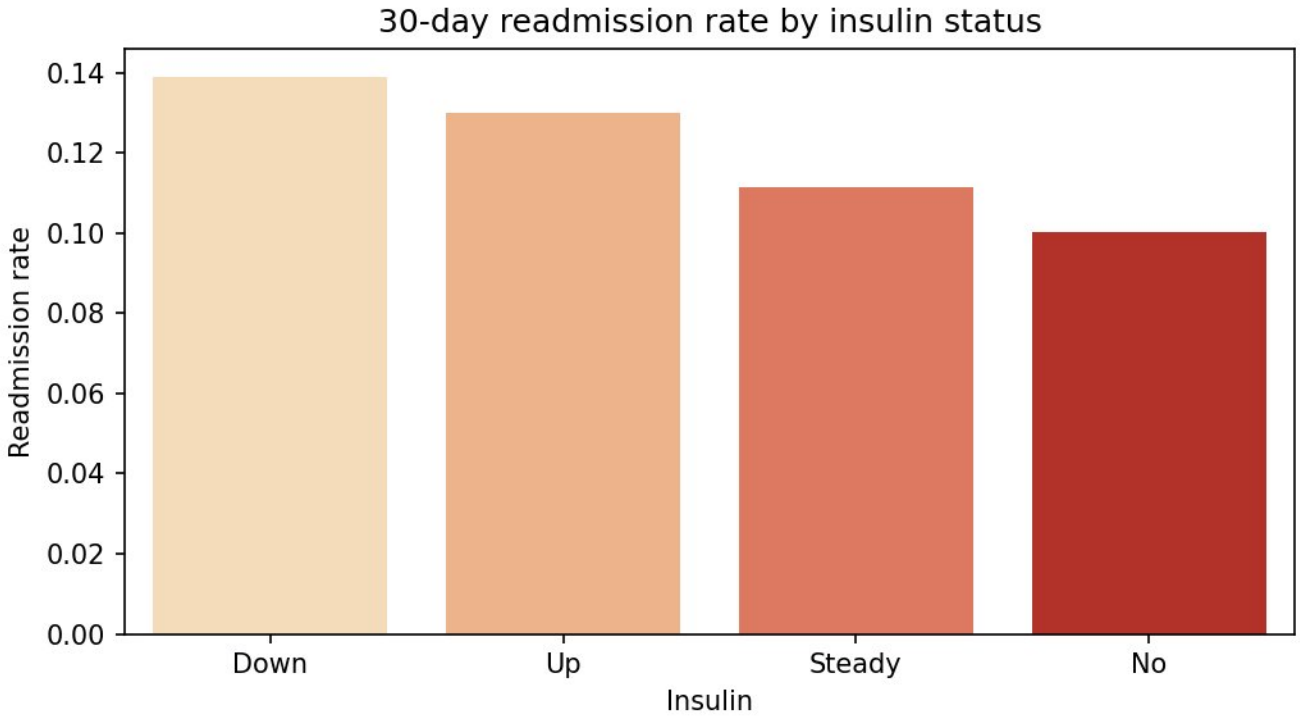


Figure 6 — Feature Scoring Across All Selection Methods (Top 20)

Heatmap showing each feature's normalized score across five selection methods: point-biserial correlation, mutual information, chi-square, Random Forest importance, and RFE. Darker cells indicate stronger signal.

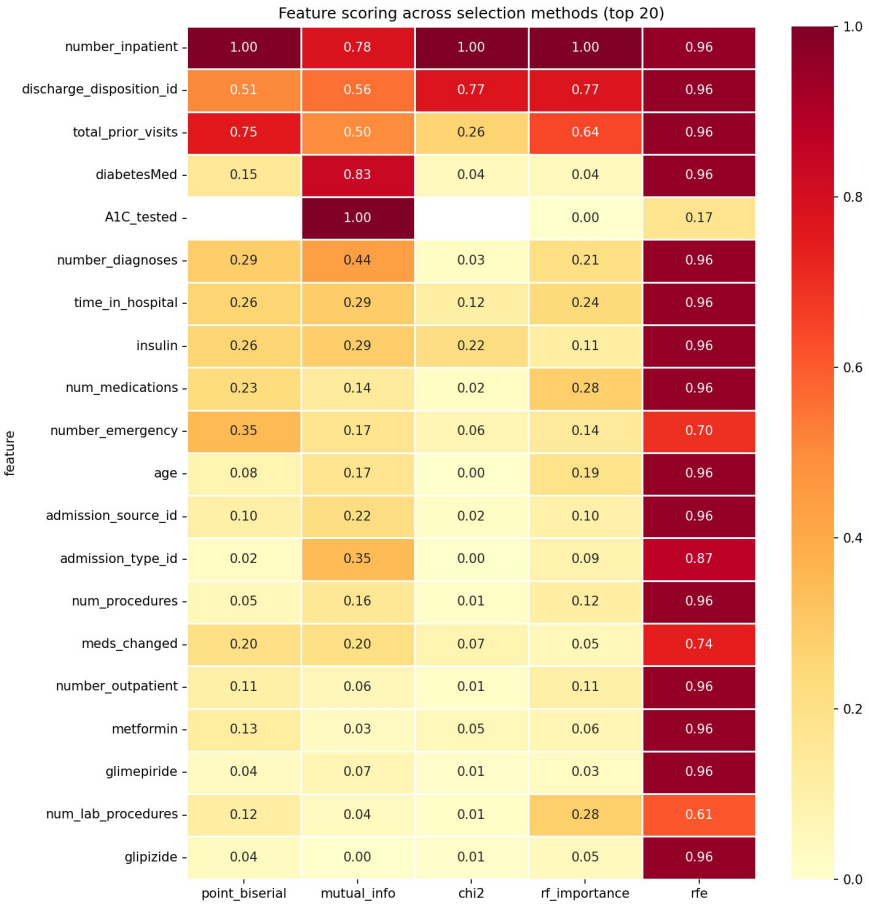


Figure 7 — Consolidated Feature Importance Ranking

Mean normalized score across all five feature selection methods. number_inpatient, discharge_disposition_id, and total_prior_visits are the clear top three. Features below 0.10 were candidates for removal.

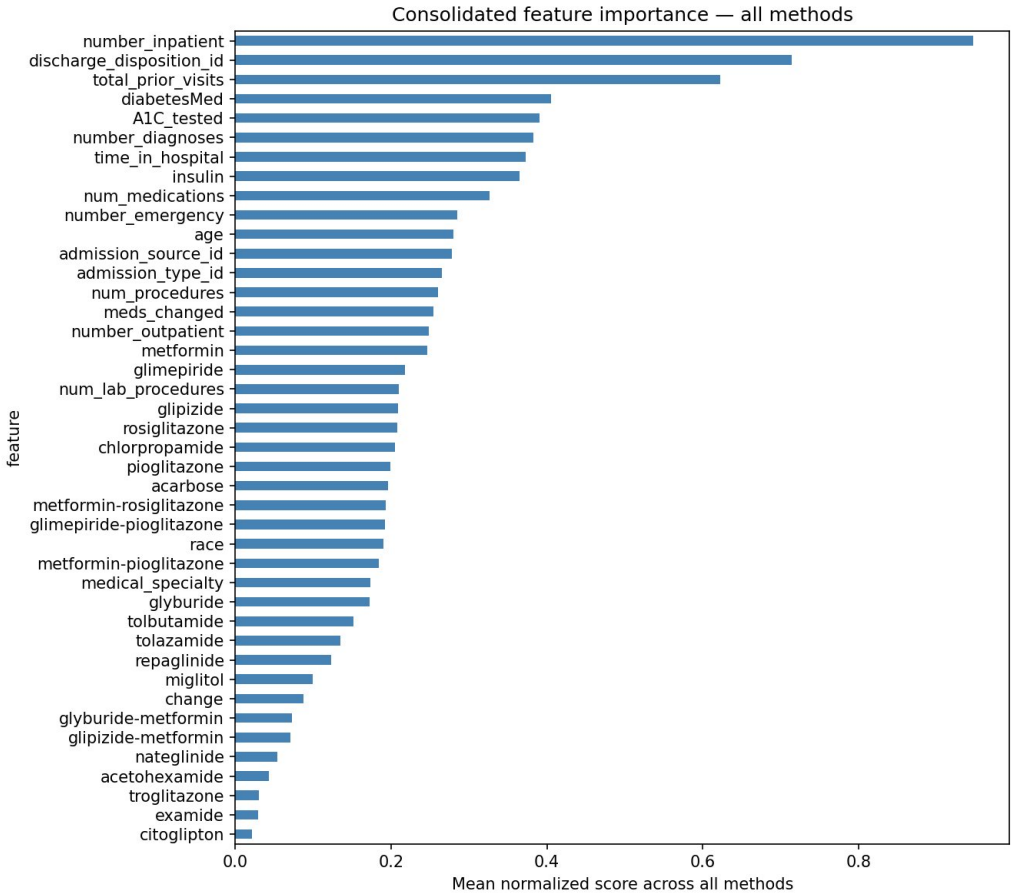


Figure 8 — Top Pairwise Feature Interactions

The strongest interaction is number_inpatient x A1C_tested (r = 0.165), meaning frequently hospitalized patients who had an A1C test ordered represent a compounded high-risk profile.

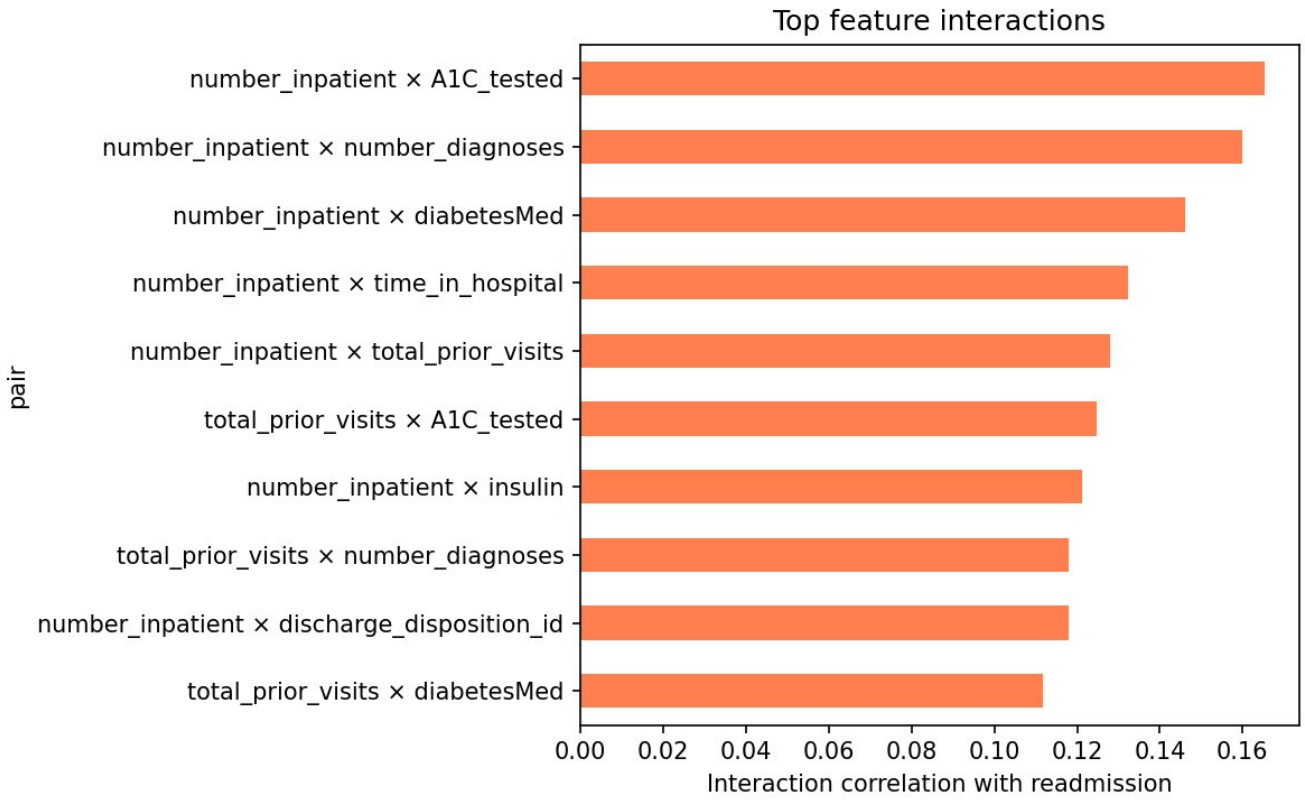


Figure 9 — ROC Curve Comparison Across All Models

XGBoost (AUC = 0.685) outperforms Random Forest (0.676) and Logistic Regression (0.659). All models substantially exceed the random baseline. The final tuned XGBoost achieved AUC = 0.690.

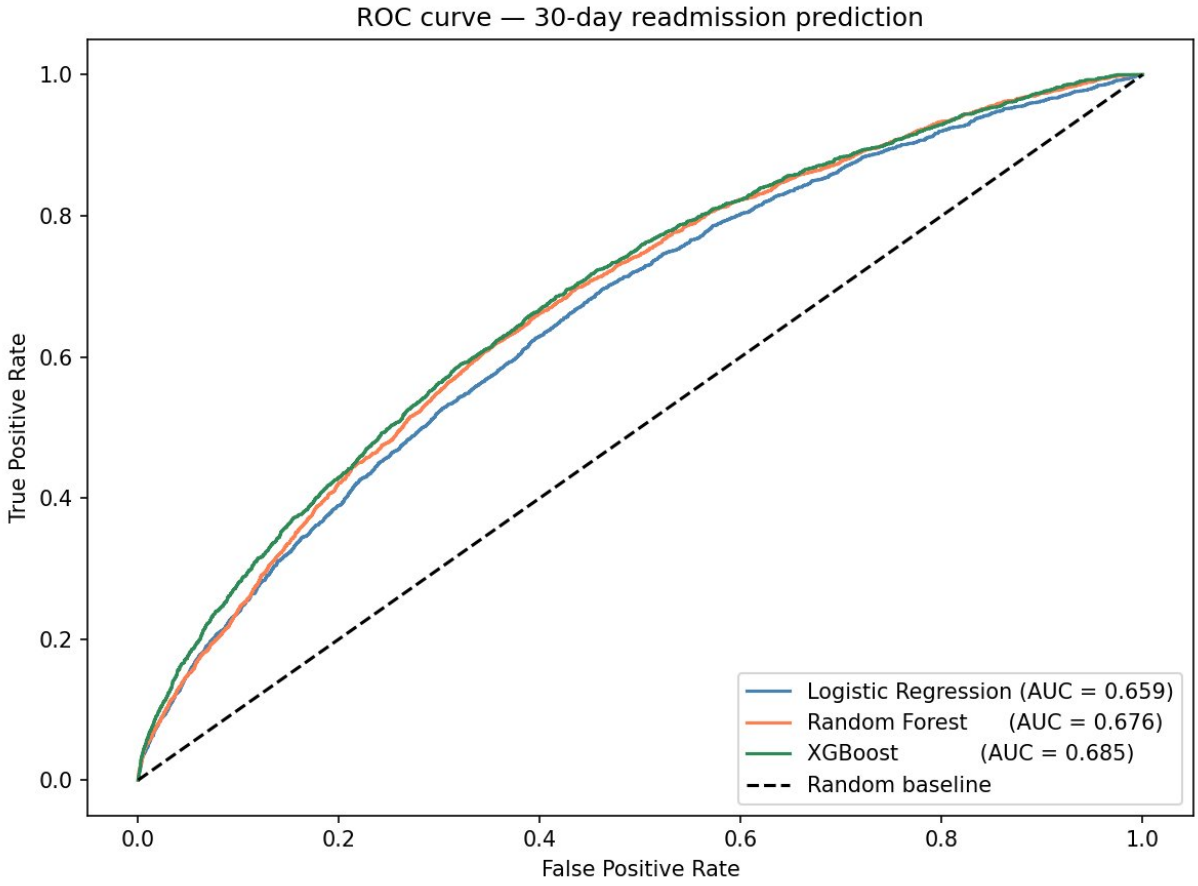


Figure 10 — XGBoost Feature Importances (Top 20)

number_inpatient dominates at 0.201, followed by discharge_disposition_id (0.157) and total_prior_visits (0.131). The engineered feature total_prior_visits ranking second validates the feature engineering approach.

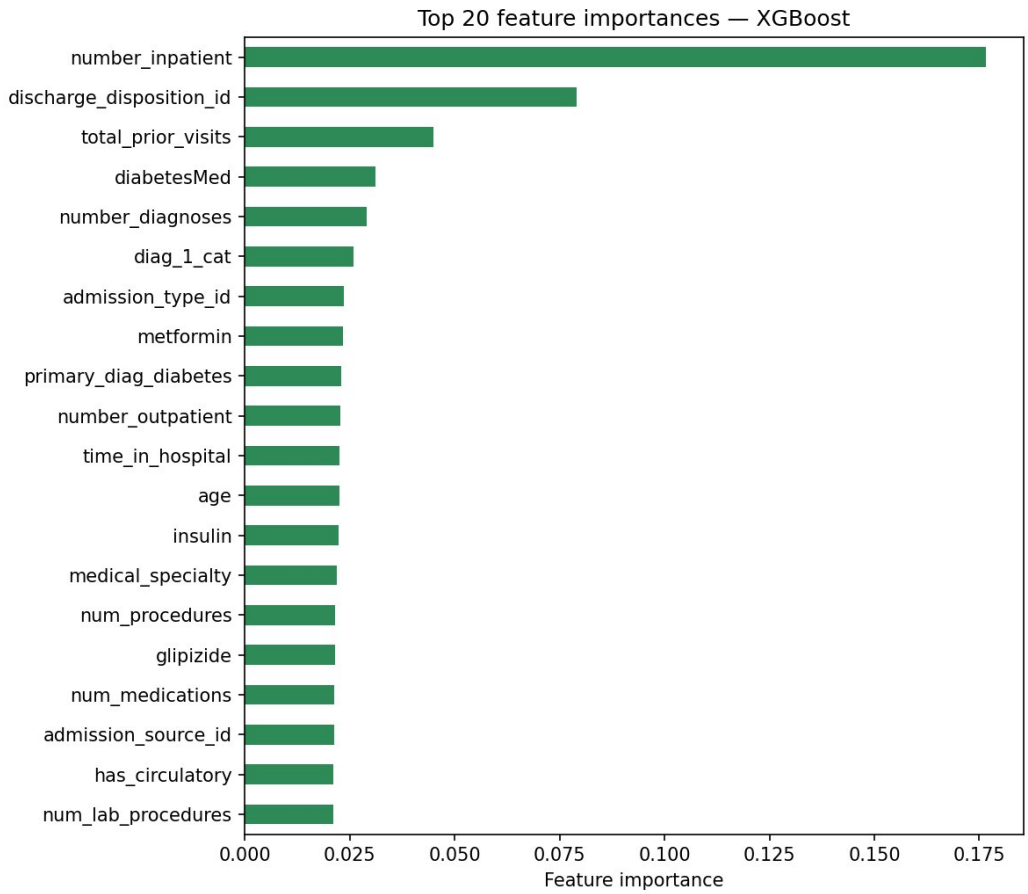


Figure 11 — Confusion Matrix (Final XGBoost Model)

On 20,354 test encounters: 1,345 readmissions correctly identified (true positives), 926 missed (false negatives), 5,906 false alarms. Recall = 59.3%, Precision = 18.6%, reflecting the 88/12 class imbalance.

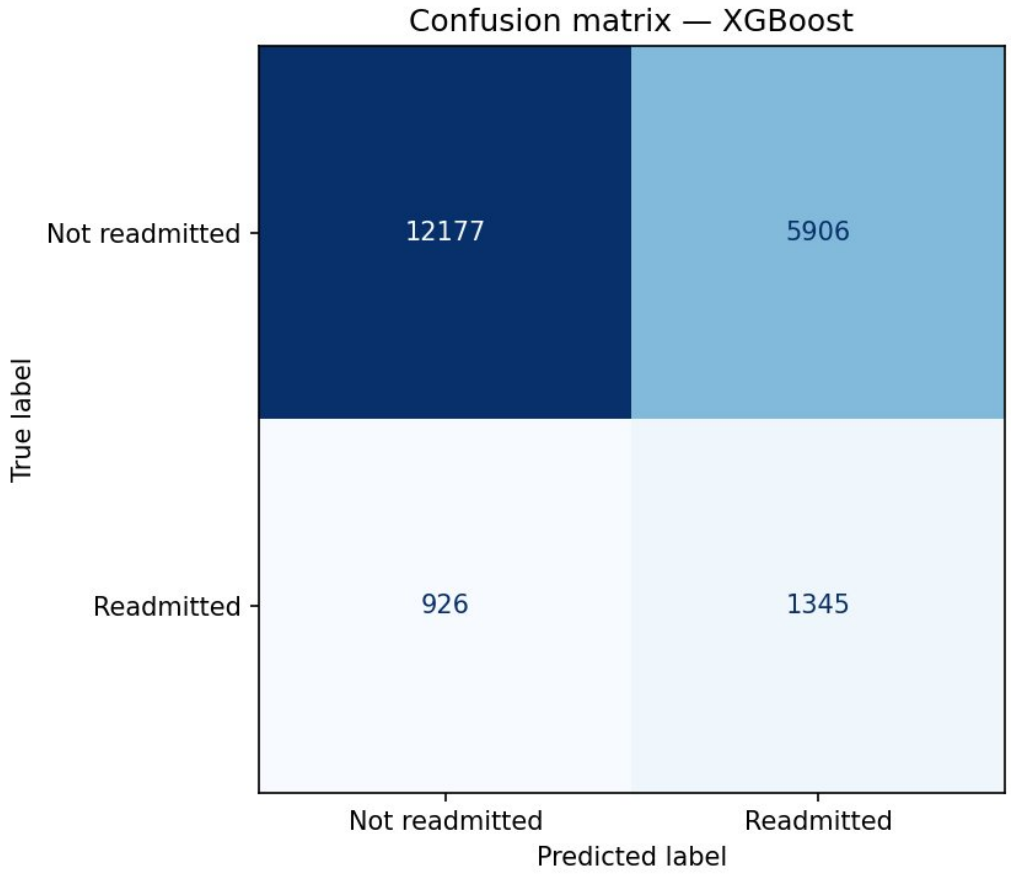
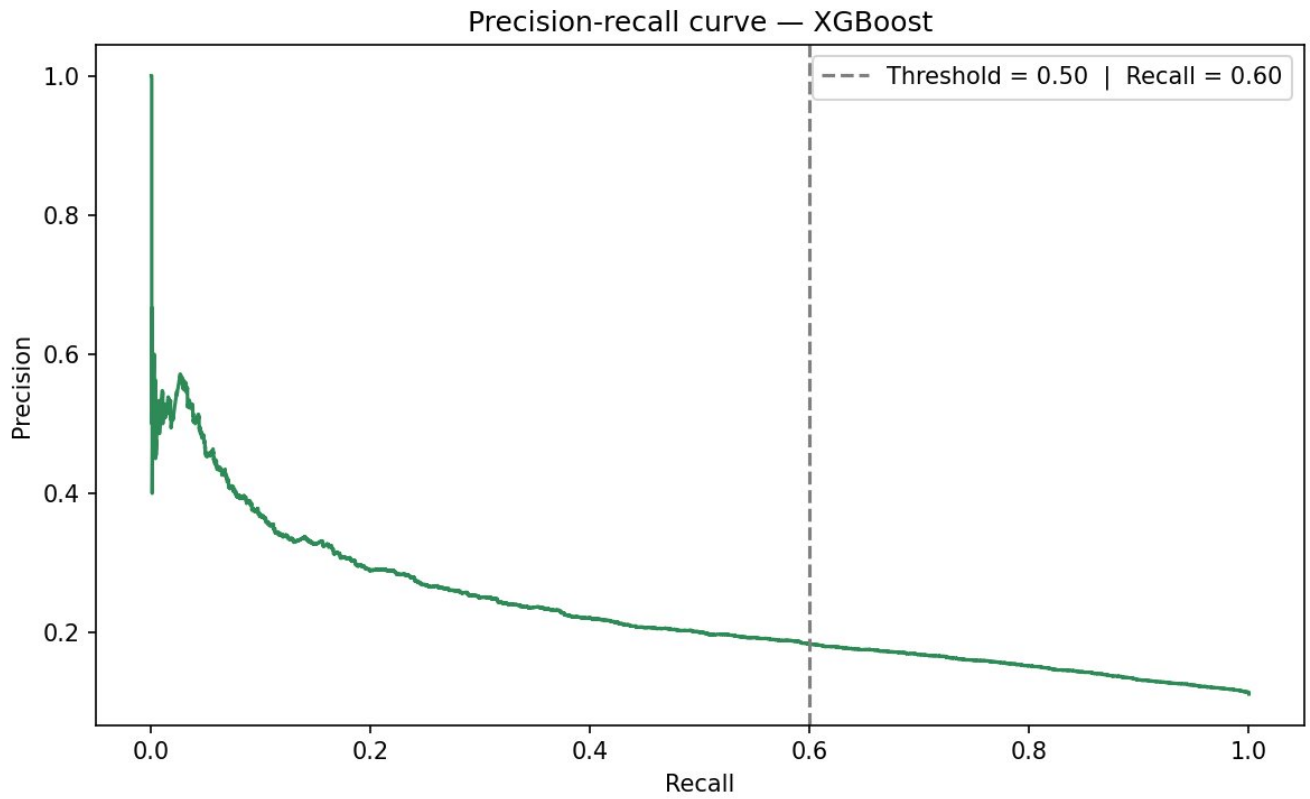


Figure 12 — Precision-Recall Curve (XGBoost)

The curve illustrates the tradeoff between catching more readmissions (higher recall) and the resulting increase in false alarms (lower precision). The dashed line marks the selected threshold at recall = 0.60.